School of Engineering

REPUBLIC POLYTECHNIC

E211 – Operations Planning II

Quiz 3 Question for Lesson 9 E-Learning

Delivery Routing

As an Operations Manager in a logistics firm, Danny needs to plan the delivery routes according to the customer's locations and demands. On a particular day, Danny needs to formulate a plan to deliver a certain product (packed in boxes) from warehouse (A) to six of its customers. The following table illustrates the distances between the warehouse (A) and the 6 customers (B to G) and their respective demands (in number of boxes).

From	To (Distance in km)							
	A	B	C	D	E	F	G	Demand
A	-	26	14	30	28	29	31	-
B	26	-	15	23	25	8	13	37
C	14	15	-	14	16	15	11	20
D	30	23	14	-	18	21	35	15
E	28	25	16	18	-	33	23	23
F	29	8	15	21	33	-	9	36
G	31	13	11	35	23	9	-	32

Determine a delivery route using the Nearest Neighbour method.

A-C-G-F-B-D-E-A

What is the total distance of the route?
14+11+9+8+23+18+28=111

Can you use another node (e.g. node B) as the starting node in the Nearest Neighbour method?
Yes, however a different solution may result

What if the vehicle can only carry 90 boxes of the product? *Select all answers that apply.*

- One route would be A -> C -> G -> F -> A
- Another route would be A-> B -> D -> E -> A
- The solution would contain 2 separate routes

  A-C-G-F-A (20+32+36=88)
  A-B-D-E-A (37+15+23=75)